

ESCUELA COLOMBIANA DE INGENIERÍA JULIO GARAVITO

IT SECURITY AND PRIVACY

INTRODUCTION TO LINUX (IMAGES)

FOCUS ON

Kali Linux

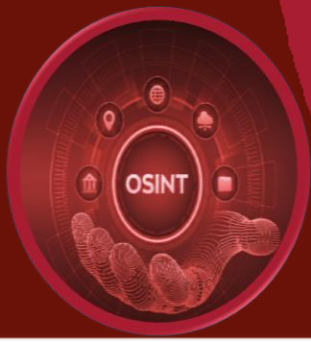
**ANDERSSON DAVID SÁNCHEZ MÉNDEZ
CRISTIAN SANTIAGO PEDRAZA RODRÍGUEZ**

PROFESSOR:

DANIEL ESTEBAN VELA LOPEZ

Place: Bogota

Date: February 03/2026

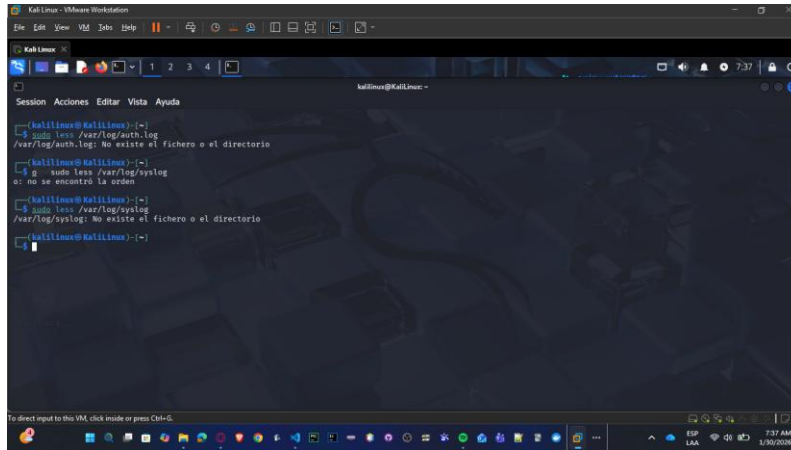



```
Kali Linux - VMware Workstation
File Edit View VM Help
Kali Linux
1 2 3 4
Session Acciones Editor Vista Ayuda
kali@kali:~$ cat /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:11:11:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
lirc:x:38:38:lirc:/var/lib/lirc:/usr/sbin/nologin
irc:x:39:39:irc:/var/lib/irc:/usr/sbin/nologin
_apt:x:42:65534:/:/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-networkd:x:998:998:systemd Network Management:/usr/sbin/nologin
dhcpd:x:100:65534:dhcp:/usr/sbin/nologin
mythtv:x:101:101:MythTV Server:/usr/sbin/nologin
tss:x:102:102:TSS Software Stack:/usr/sbin/nologin
strongswan:x:103:65534:/var/lib/strongswan:/usr/sbin/nologin
systemd-timesyncd:x:991:991:systemd Time Synchronization:/usr/sbin/nologin
_aptlib:x:104:104:/var/lib/aptlib:/usr/sbin/nologin
sddm:x:105:65534:/run/sddm:/usr/sbin/nologin
messagebus:x:1000:1000:Message Bus:/nonexistent:/usr/sbin/nologin
tcpdump:x:106:107:/nonexistent:/usr/sbin/nologin
```

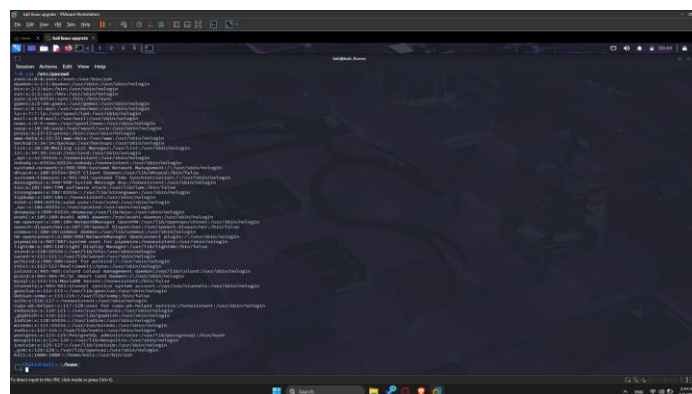
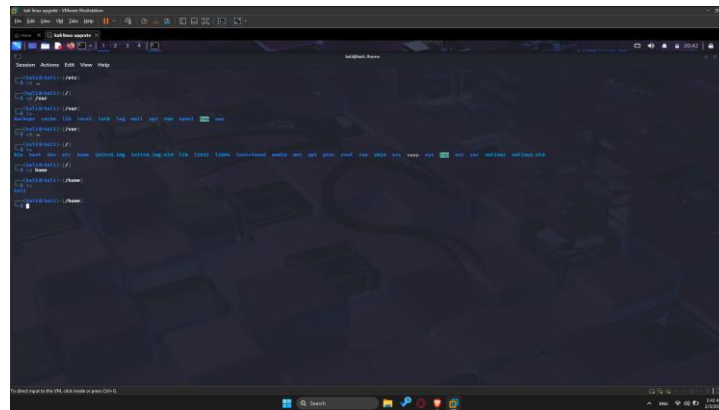
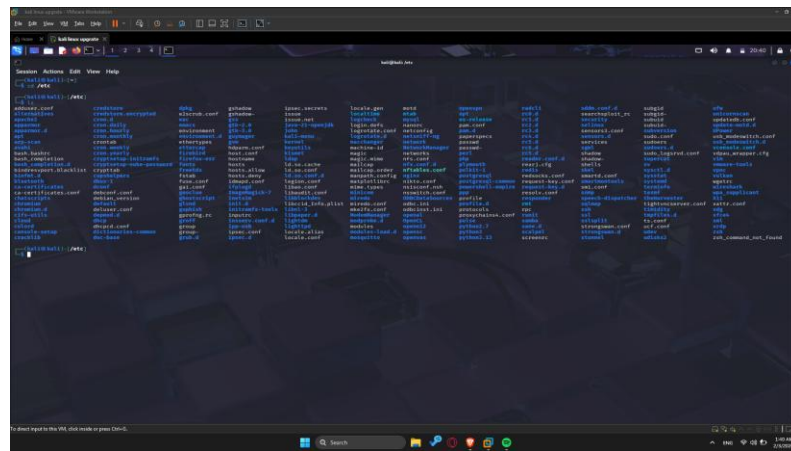
```
Kali Linux - VMware Workstation
File Edit View VM Help
Kali Linux
1 2 3 4
Session Acciones Editor Vista Ayuda
kali@kali:~$ cat /etc/passwd
_rpc:x:100:65534:/run/rpcbind:/usr/sbin/nologin
redis:x:109:108:/var/lib/redis:/usr/sbin/nologin
mosquitto:x:110:111:/var/lib/mosquitto:/usr/sbin/nologin
redis:x:111:111:/var/lib/redis:/usr/sbin/nologin
stunnel:x:100:981:stunnel service system account:/var/run/stunnel4:/usr/sbin/nologin
sshd:x:100:65534:ssh:/run/ssh:/usr/sbin/nologin
dmesd:x:999:65534:dmesd:/usr/lib/dmcc:/usr/sbin/nologin
Debian-smp:x:112:112:/var/lib/smp:/bin/false
sblinx:x:113:113:/nonexistent:/usr/sbin/nologin
avahi:x:114:114:Avahi mDNS daemon:/usr/sbin/nologin
speech-dispatcher:x:115:79:Speech Dispatcher:/usr/bin/speech-dispatcher:/bin/false
_gnss:x:116:116:/var/lib/gnss:/usr/sbin/nologin
postgre:x:117:118:PostgreSQL administrator:/var/lib/postgresql:/bin/bash
colman:x:118:118:colman:/var/lib/colman:/usr/sbin/nologin
cups-pk-helper:x:119:121:User for cups-pk-helper service:/nonexistent:/usr/sbin/nologin
m-openssh:x:120:121:OpenSSH OpenSSH plugin:/usr/sbin/nologin
inetsix:x:121:122:/var/lib/inetsix:/usr/sbin/nologin
pipewire:x:198:198:system user for pipewire:/nonexistent:/usr/sbin/nologin
m-openssh:x:199:199:OpenSSH OpenSSH plugin:/usr/sbin/nologin
geoclue:x:122:122:/var/lib/geoclue:/usr/sbin/nologin
lightdm:x:123:124:LightDM Display Manager:/usr/lib/lightdm:/bin/false
statd:x:124:65534:/var/lib/rfss:/usr/sbin/nologin
samed:x:125:125:/var/lib/samed:/usr/sbin/nologin
polkit:x:100:981:User for polkit:/usr/sbin/nologin
rktlib:x:126:126:rktlib:/usr/lib/rktlib:/usr/sbin/nologin
colord:x:100:981:colord colour management daemon:/usr/lib/colord:/usr/sbin/nologin
pcscd:x:100:981:PC/SC Smart Card Daemon:/usr/sbin/nologin
kali@kali:~$ cat /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
```

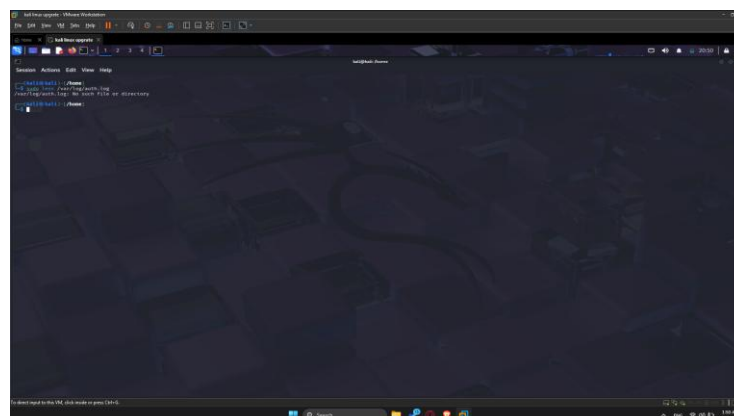
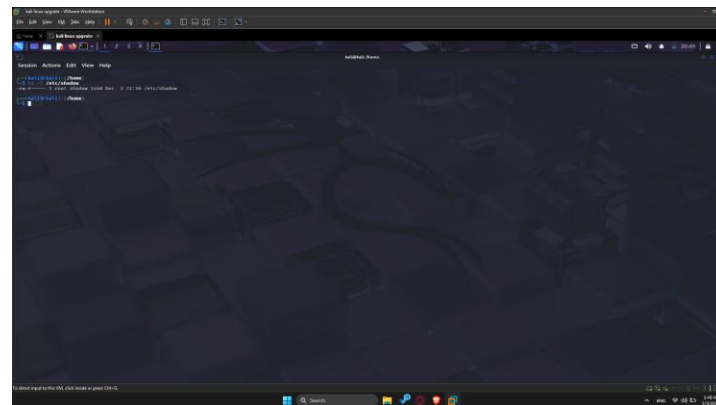
```
Kali Linux - VMware Workstation
File Edit View VM Help
Kali Linux
1 2 3 4
Session Acciones Editor Vista Ayuda
kali@kali:~$ cat /etc/shadow
cat: /etc/shadow: Permisos denegados
kali@kali:~$ cat /etc/shadow
[sudo] contraseña para kali@kali:
root:!:20474:0:99999:7:::
daemon:!:20474:0:99999:7:::
bin:!:20474:0:99999:7:::
sys:!:20474:0:99999:7:::
sync:!:20474:0:99999:7:::
games:!:20474:0:99999:7:::
man:!:20474:0:99999:7:::
lp:!:20474:0:99999:7:::
mail:!:20474:0:99999:7:::
news:!:20474:0:99999:7:::
uucp:!:20474:0:99999:7:::
proxy:!:20474:0:99999:7:::
www-data:!:20474:0:99999:7:::
backup:!:20474:0:99999:7:::
lirc:!:20474:0:99999:7:::
irc:!:20474:0:99999:7:::
nobody:!:20474:0:99999:7:::
systemd-networkd:!:20474:0:99999:7:::
dhcpd:!:20474:0:99999:7:::
mythtv:!:20474:0:99999:7:::
tss:!:20474:0:99999:7:::
strongswan:!:20474:0:99999:7:::
```

```
Kali Linux - VMware Workstation
File Edit View VM Help
Kali Linux
1 2 3 4
Session Acciones Editor Vista Ayuda
kali@kali:~$ cat /etc/shadow
cat: /etc/shadow: Permisos denegados
kali@kali:~$ cat /etc/shadow
[sudo] contraseña para kali@kali:
root:!:20474:0:99999:7:::
daemon:!:20474:0:99999:7:::
bin:!:20474:0:99999:7:::
sys:!:20474:0:99999:7:::
sync:!:20474:0:99999:7:::
games:!:20474:0:99999:7:::
man:!:20474:0:99999:7:::
lp:!:20474:0:99999:7:::
mail:!:20474:0:99999:7:::
news:!:20474:0:99999:7:::
uucp:!:20474:0:99999:7:::
proxy:!:20474:0:99999:7:::
www-data:!:20474:0:99999:7:::
backup:!:20474:0:99999:7:::
lirc:!:20474:0:99999:7:::
irc:!:20474:0:99999:7:::
nobody:!:20474:0:99999:7:::
systemd-networkd:!:20474:0:99999:7:::
dhcpd:!:20474:0:99999:7:::
mythtv:!:20474:0:99999:7:::
tss:!:20474:0:99999:7:::
strongswan:!:20474:0:99999:7:::
```

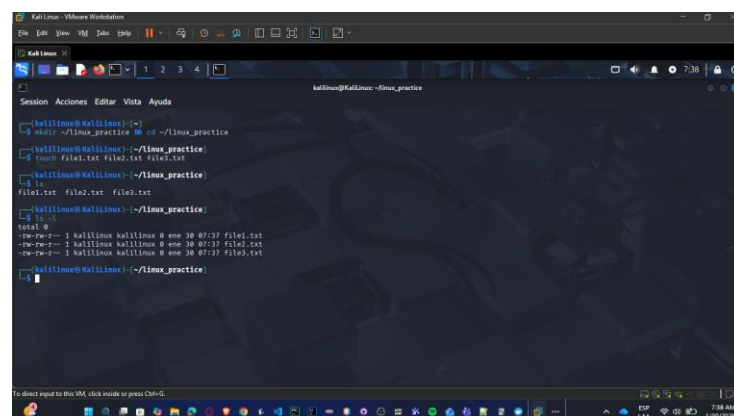


Cristian's images:





Andersson's images:




```
Kali Linux - VMware Workstation
File Edit View VM Tools Help
Kali Linux
Session Acciones Editor Vista Ayuda
kali@kali:~$ ping 8.8.8.8 -c 1
ping 8.8.8.8: 6544 2356 pts/0  S+  07:45  0:00  grep --color=auto ssh
kali@kali:~$
```

```
Kali Linux - VMware Workstation
File Edit View VM Tools Help
Kali Linux
Session Acciones Editor Vista Ayuda
kali@kali:~$ ping 8.8.8.8 -c 1
ping 8.8.8.8: 6964 2700 pts/0  SN  07:46  0:00  ping 8.8.8.8
kali@kali:~$ ping 8.8.8.8 -c 1
ping 8.8.8.8: 7112 2300 pts/0  S+  07:46  0:00  grep --color=auto ssh
kali@kali:~$
```

```
Kali Linux - VMware Workstation
File Edit View VM Tools Help
Kali Linux
Session Acciones Editor Vista Ayuda
kali@kali:~$ systemctl status ssh
Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
Active: inactive (dead)
Docs: man:ssh(8)
      man:ssh_config(5)

kali@kali:~$ service cron status
cron.service - Regular background program processing daemon
Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
Active: active (running) since Fri 2020-01-30 07:35:29 -05; 12min ago
Invocation: c58ff1fbd1744ecb42f78d97847e15
Docs: man:cron(8)
Main PID: 600 (cron)
Tasks: 2 (limit: 2073)
Memory: 696K (peak: 2.5M)
CPU: 50ms
CGroup: /system.slice/cron.service
└─600 /usr/sbin/cron -f

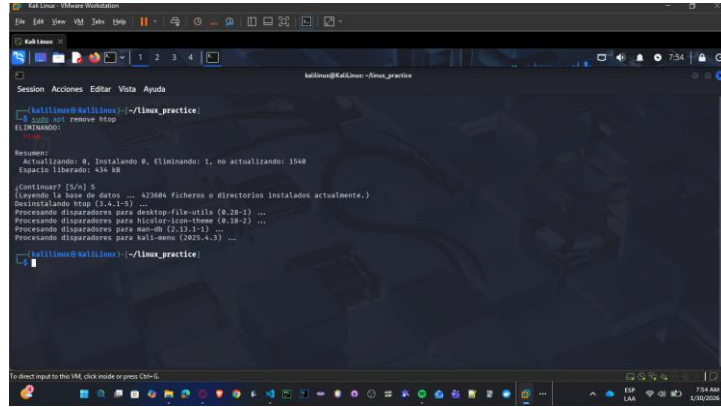
ene 30 07:35:29 kali Linux systemd[1]: Started cron.service - Regular background program processing daemon.
ene 30 07:35:29 kali Linux cron[600]: (CRON) INFO (pidfile fd = 3)
ene 30 07:35:29 kali Linux cron[600]: (CRON) INFO (Running 87 boot jobs)
ene 30 07:39:01 kali Linux CRON[3338]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
ene 30 07:39:01 kali Linux CRON[3338]: (root) CMD ( [ -x /usr/lib/php/sessionclean ] && if [ ! -d /run/systemd/system ]; then /usr/lib/php/sessionclean; fi)
ene 30 07:39:01 kali Linux CRON[3338]: pam_unix(cron:session): session closed for user root
ene 30 07:39:01 kali Linux CRON[3338]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
ene 30 07:45:01 kali Linux CRON[6359]: (root) CMD (command -v debian-sa1 > /dev/null 06 debian-sa1 1 1)
ene 30 07:45:01 kali Linux CRON[6359]: pam_unix(cron:session): session closed for user root
```

```
Kali Linux - VMware Workstation
File Edit View VM Tools Help
Kali Linux
Session Acciones Editor Vista Ayuda
kali@kali:~$ systemctl status ssh
Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
Active: inactive (dead)
Docs: man:ssh(8)
      man:ssh_config(5)

kali@kali:~$ service cron status
cron.service - Regular background program processing daemon
Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
Active: active (running) since Fri 2020-01-30 07:35:29 -05; 12min ago
Invocation: c58ff1fbd1744ecb42f78d97847e15
Docs: man:cron(8)
Main PID: 600 (cron)
Tasks: 2 (limit: 2073)
Memory: 696K (peak: 2.5M)
CPU: 50ms
CGroup: /system.slice/cron.service
└─600 /usr/sbin/cron -f

ene 30 07:35:29 kali Linux systemd[1]: Started cron.service - Regular background program processing daemon.
ene 30 07:35:29 kali Linux cron[600]: (CRON) INFO (pidfile fd = 3)
ene 30 07:35:29 kali Linux cron[600]: (CRON) INFO (Running 87 boot jobs)
ene 30 07:39:01 kali Linux CRON[3338]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
ene 30 07:39:01 kali Linux CRON[3338]: (root) CMD ( [ -x /usr/lib/php/sessionclean ] && if [ ! -d /run/systemd/system ]; then /usr/lib/php/sessionclean; fi)
ene 30 07:39:01 kali Linux CRON[3338]: pam_unix(cron:session): session closed for user root
ene 30 07:39:01 kali Linux CRON[3338]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
ene 30 07:45:01 kali Linux CRON[6359]: (root) CMD (command -v debian-sa1 > /dev/null 06 debian-sa1 1 1)
ene 30 07:45:01 kali Linux CRON[6359]: pam_unix(cron:session): session closed for user root

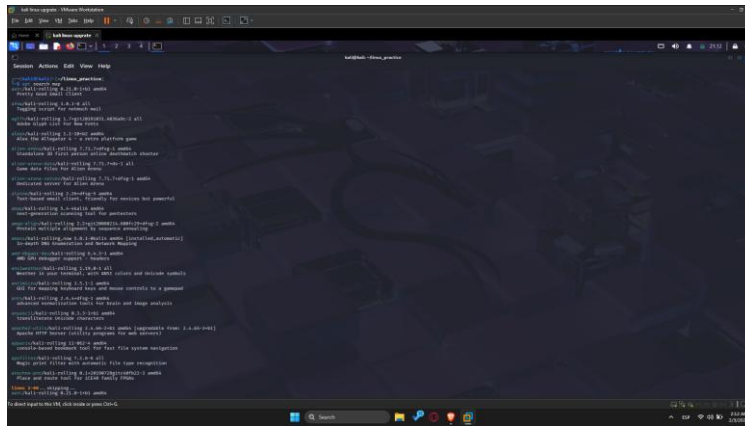
kali@kali:~$
```

```

Kali Linux - VMware Workstation
Kali Linux X
Session Acciones Editar Vista Ayuda
kali@kali:~/linux_practice
~
kali@kali:~/linux_practice$ rm -rf http
ELIMINANDO:
http
Resumen:
Actualizando: 0, Instalando 0, Eliminando: 1, no actualizando: 1540
Espacio liberado: 434 kB
(Continuar? [S/n]) S
Leyendo la base de datos ... 423604 ficheros o directorios instalados actualmente.)
Desinstalando htop (3.4.1-3) ...
Procesando disparadores para desktop-file-utils (0.26-1) ...
Procesando disparadores para hicolor-icon-theme (0.16-2) ...
Procesando disparadores para menu (2.13.1-3) ...
Procesando disparadores para kali-menu (2025.4.3) ...
kali@kali:~/linux_practice$
  
```

Cristian's images:



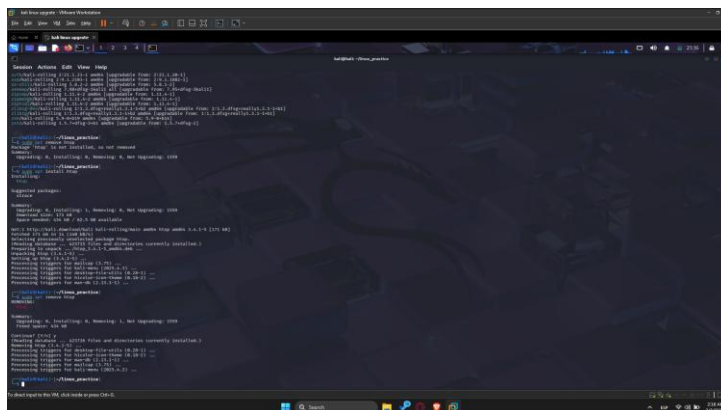
```

Kali Linux - VMware Workstation
Kali Linux X
Session Acciones Editar Vista Ayuda
kali@kali:~/linux_practice
~
kali@kali:~/linux_practice$ rm -rf http
ELIMINANDO:
http
Resumen:
Actualizando: 0, Instalando 0, Eliminando: 1, no actualizando: 1540
Espacio liberado: 434 kB
(Continuar? [S/n]) S
Leyendo la base de datos ... 423604 ficheros o directorios instalados actualmente.)
Desinstalando htop (3.4.1-3) ...
Procesando disparadores para desktop-file-utils (0.26-1) ...
Procesando disparadores para hicolor-icon-theme (0.16-2) ...
Procesando disparadores para menu (2.13.1-3) ...
Procesando disparadores para kali-menu (2025.4.3) ...
kali@kali:~/linux_practice$
  
```



```

Kali Linux - VMware Workstation
Kali Linux X
Session Acciones Editar Vista Ayuda
kali@kali:~/linux_practice
~
kali@kali:~/linux_practice$ rm -rf http
ELIMINANDO:
http
Resumen:
Actualizando: 0, Instalando 0, Eliminando: 1, no actualizando: 1540
Espacio liberado: 434 kB
(Continuar? [S/n]) S
Leyendo la base de datos ... 423604 ficheros o directorios instalados actualmente.)
Desinstalando htop (3.4.1-3) ...
Procesando disparadores para desktop-file-utils (0.26-1) ...
Procesando disparadores para hicolor-icon-theme (0.16-2) ...
Procesando disparadores para menu (2.13.1-3) ...
Procesando disparadores para kali-menu (2025.4.3) ...
kali@kali:~/linux_practice$
  
```

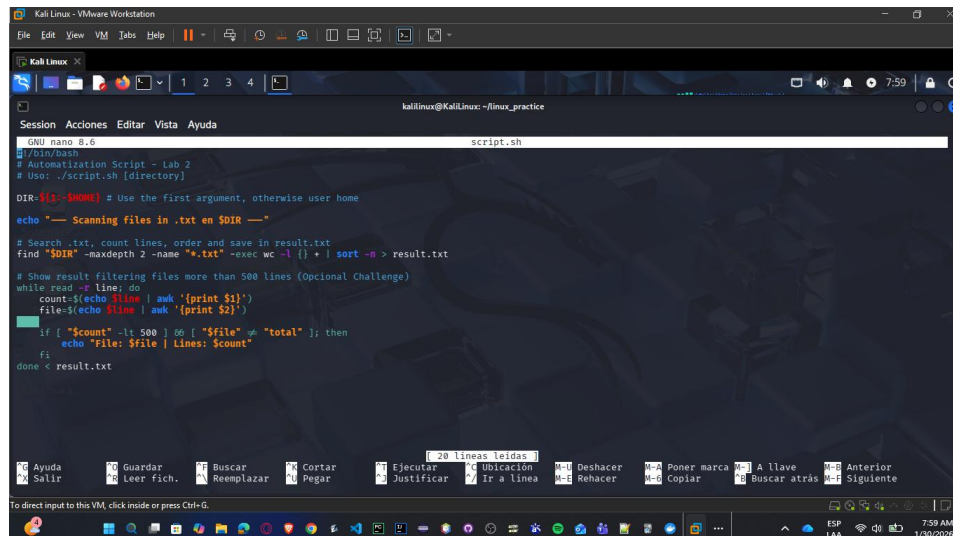


```

Kali Linux - VMware Workstation
Kali Linux X
Session Acciones Editar Vista Ayuda
kali@kali:~/linux_practice
~
kali@kali:~/linux_practice$ rm -rf http
ELIMINANDO:
http
Resumen:
Actualizando: 0, Instalando 0, Eliminando: 1, no actualizando: 1540
Espacio liberado: 434 kB
(Continuar? [S/n]) S
Leyendo la base de datos ... 423604 ficheros o directorios instalados actualmente.)
Desinstalando htop (3.4.1-3) ...
Procesando disparadores para desktop-file-utils (0.26-1) ...
Procesando disparadores para hicolor-icon-theme (0.16-2) ...
Procesando disparadores para menu (2.13.1-3) ...
Procesando disparadores para kali-menu (2025.4.3) ...
kali@kali:~/linux_practice$
  
```

5. Automation with Bash

Andersson's images:



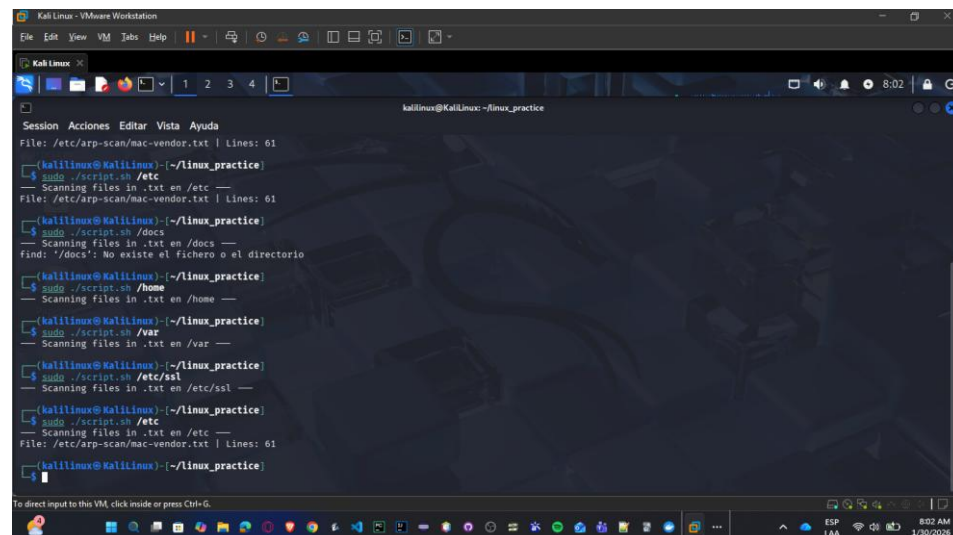
```
GNU nano 8.6 script.sh
#!/bin/bash
# Automatization Script - Lab 2
# User: ./script.sh [directory]

DIR=${1:-$HOME} # Use the first argument, otherwise user home

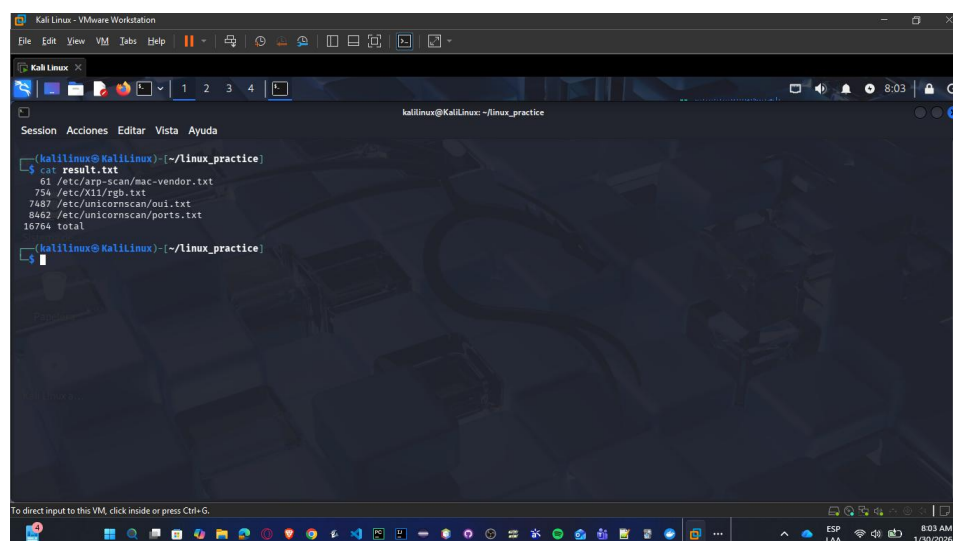
echo "— Scanning files in .txt en $DIR —"

# Search .txt, count lines, order and save in result.txt
find "$DIR" -maxdepth 2 -name "*.txt" -exec wc -l {} + | sort -n > result.txt

# Show result filtering files more than 500 lines (Optional Challenge)
while read -r line; do
    count=$(echo $line | awk '{print $1}')
    file=$(echo $line | awk '{print $2}')
    if [ "$count" -lt 500 ] || [ "$file" = "total" ]; then
        echo "File: $file | Lines: $count"
    fi
done < result.txt
```



```
File: /etc/arp-scan/mac-vendor.txt | Lines: 61
(kalilinux@kalilinux) ~/linux_practice
$ sudo ./script.sh /etc
— Scanning files in .txt en /etc —
File: /etc/arp-scan/mac-vendor.txt | Lines: 61
(kalilinux@kalilinux) ~/linux_practice
$ sudo ./script.sh /docs
— Scanning files in .txt en /docs —
find: /docs: No existe el fichero o el directorio
(kalilinux@kalilinux) ~/linux_practice
$ sudo ./script.sh /home
— Scanning files in .txt en /home —
(kalilinux@kalilinux) ~/linux_practice
$ sudo ./script.sh /var
— Scanning files in .txt en /var —
(kalilinux@kalilinux) ~/linux_practice
$ sudo ./script.sh /etc/ssl
— Scanning files in .txt en /etc/ssl —
(kalilinux@kalilinux) ~/linux_practice
$ sudo ./script.sh /etc
— Scanning files in .txt en /etc —
File: /etc/arp-scan/mac-vendor.txt | Lines: 61
(kalilinux@kalilinux) ~/linux_practice
$
```



```
(kalilinux@kalilinux) ~/linux_practice
$ cat result.txt
61 /etc/arp-scan/mac-vendor.txt
754 /etc/x11/rgb.txt
7487 /etc/unicornsand/oui.txt
8462 /etc/unicornsand/ports.txt
16764 total
(kalilinux@kalilinux) ~/linux_practice
$
```

Cristian's images:

